Lecture 31.01: Introduction to Unsupervised Machine Learning

Status Complete

Topic Theory & Math

Notes Hinglish

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📌 Topic Overview

Comprehensive introduction to Unsupervised Machine Learning. Unlike supervised learning, unsupervised learning operates on unlabeled feature sets to discover latent structures, clusters, dimensional reductions, and anomalies.

📋 Key Learning Objectives

- Fundamental contrast: Supervised Learning (X, y) vs Unsupervised Learning (X only).
- The 3 Primary Branches: Clustering, Dimensionality Reduction, and Anomaly Detection.
- Real-World Bank ATM Case Study: Customer spending segmentation (Balance vs Cash Withdrawals).
- Industry Applications: Market basket analysis, fraud detection, and recommendation engines.

🔍 In-Depth Conceptual Breakdown

- **Supervised vs Unsupervised Learning**: Supervised learning mein hamare pass inputs X ke sath target output labels y hote the (jaise Spam/Ham ya House Price). Lekin **Unsupervised Learning** mein koi target label y nahi hota! Model ko sirf input features X diye jaate hain aur use khud data ke andar ke hidden patterns aur groupings dhoondhne hote hain.

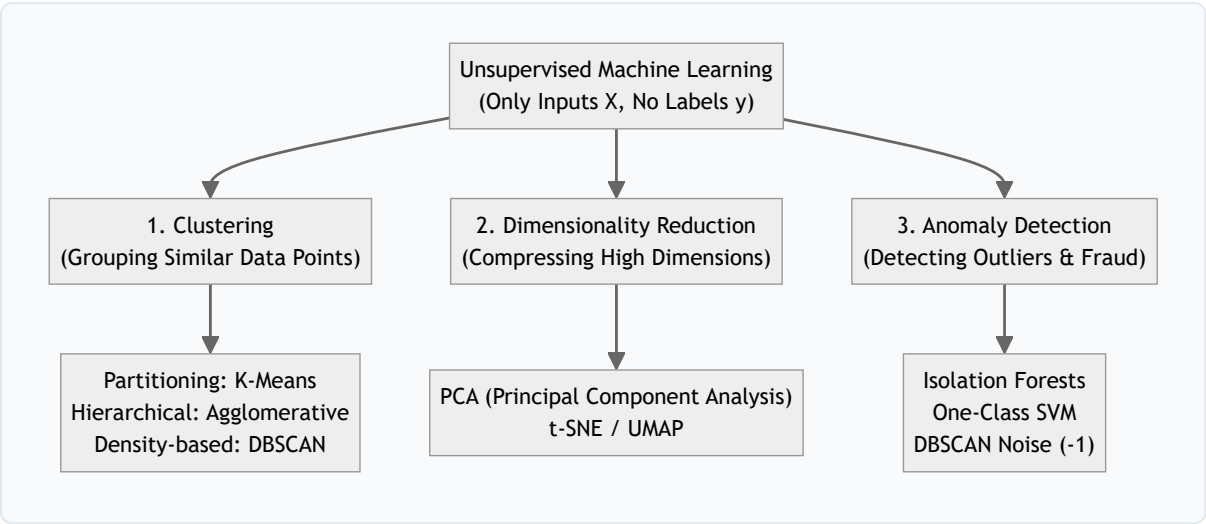
- **Pehli Main Branch: Clustering**: Unlabeled data points ko unki similarities ke basis par alag-alag groups (clusters) mein divide karna. Jaise customer segmentation ya document topic classification.

- **Doosri Main Branch: Dimensionality Reduction**: Jab dataset mein bohot saare features (jaise 100+ columns) hote hain, toh unhe 2D ya 3D mein compress karna bina zaroori information khoe (jaise **PCA**, **t-SNE**).
- **Teesri Main Branch: Anomaly Detection**: Normal patterns se alag hatkar aane wale unusual ya fraudulent data points ko pehchanna (jaise Credit Card Fraud ya Server Defect detection).
- **Classroom Bank ATM Example**: Ma'am ne board par bank customer transactions ka example diya jisme do features the: `Feature 1 = Daily Balance` aur `Feature 2 = Cash Withdrawals`. Bank apne customers ko 2 clusters mein segment karta hai: high balance/low withdrawals wale users ko investment offers bhejna, aur low balance/high withdrawals wale users ko personal loan offers bhejna!
- **Pure Theory Lecture**: Is lecture mein ma'am ne conceptual foundations samjhaye hain, isliye is module mein koi code nahi hai.

📋 Supervised vs Unsupervised Learning Comparison Matrix

Feature / Dimension	Supervised Machine Learning	Unsupervised Machine Learning
Training Data	Features X + Ground Truth Labels y	Only Features X (Unlabeled Data)
Objective	Predict label y for unseen sample x^*	Discover hidden patterns, clusters, manifolds
Evaluation	Ground truth metrics (Accuracy, MSE, ROC-AUC)	Internal geometric metrics (Inertia, Silhouette)
Primary Tasks	Classification, Regression	Clustering, Dim-Reduction, Anomaly Detection
Feedback Mechanism	Error feedback via actual label $(y - \hat{y})$	No external feedback (Self-guided discovery)

📊 Unsupervised Machine Learning Taxonomy



🔢 Mathematical Formulation

Data representation in unsupervised learning:

$$\mathcal{D} = \{\mathbf{x}_1, \mathbf{x}_2, \ldots, \mathbf{x}_N\}, \quad \text{where } \mathbf{x}_i \in \mathbb{R}^D$$

Objective of clustering into K disjoint partitions $\{S_1, S_2, \ldots, S_K\}$:

$$\bigcup_{k=1}^{K} S_k = \mathcal{D}, \quad S_j \cap S_k = \emptyset \quad (\forall j \neq k)$$

📌 Note on Code

This is a pure conceptual, mathematical, and blackboard theory lecture. In accordance with course guidelines, no synthetic code is appended.